

Detecting Explicit and *Subtle* Toxicity

Bangla, English and Banglish social media — macro F1 0.77514

HerWILL × UAP Safe Social Media Datathon 2026

Team FatWiz ·

The problem

Classify each social-media comment:

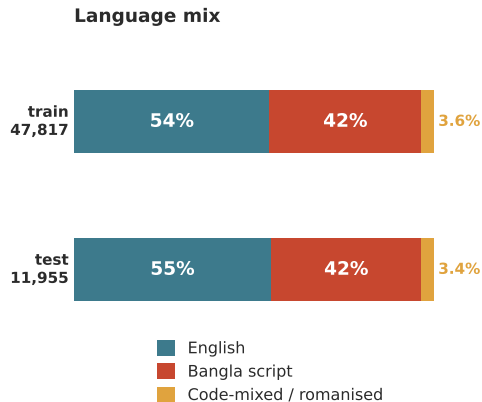
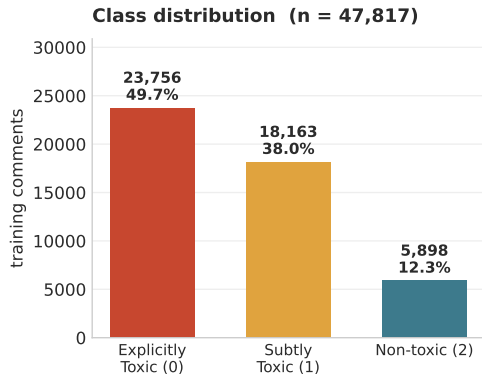
- 0** Explicitly Toxic — open abuse, slurs, threats
- 1** Subtly Toxic — sarcasm, mockery, coded language
- 2** Non-toxic — opinion, news, criticism of ideas

Setting

- **47,817** train / **11,955** test — raw text, no metadata
- Bangla, English and Banglish in one corpus

Metric: macro F1

We measured the data before choosing a model



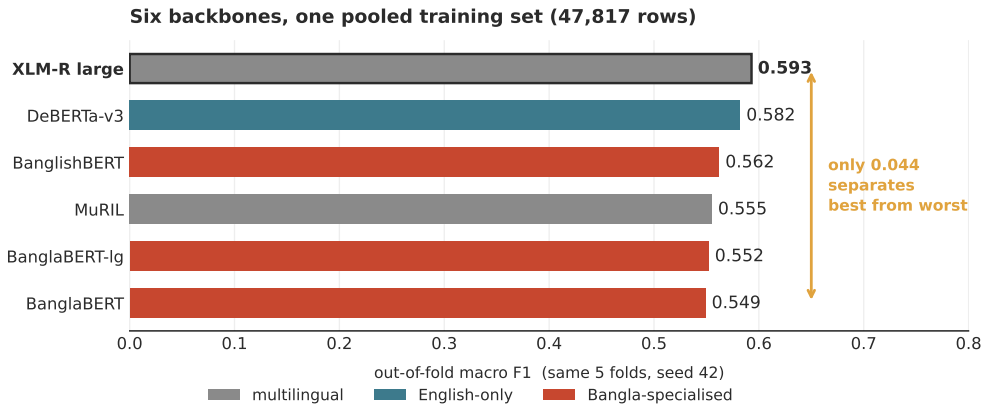
- Train and test agree to **0.4pp** $\Rightarrow$ **no distribution shift**

- Median comment **47 characters** (p90 = 129)

Preprocessing: every step has a number behind it

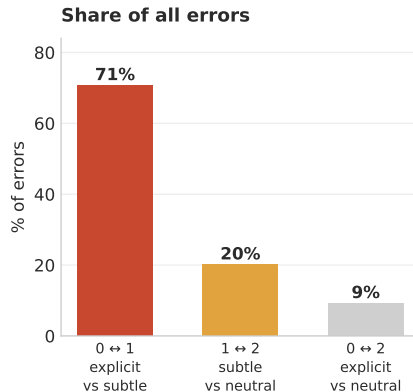
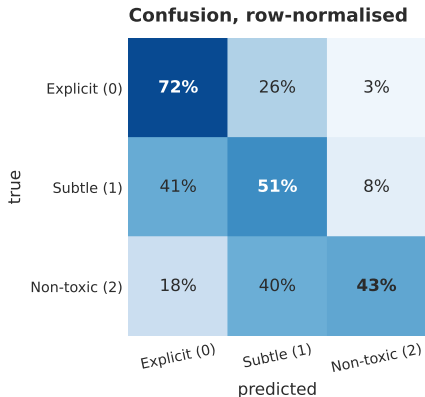
Action	Measured	Why
FIXED	16.2%	rows not Unicode NFC — the same Bengali word tokenises two ways
KEPT	9.1%	emoji · 8.5% repeated punctuation · 1.2% censored spellings — <i>tone is the label</i> for subtle toxicity
63 identical texts carry conflicting labels		Deduplicated across the merge — 669 duplicate texts removed.

Six backbones on the full training set

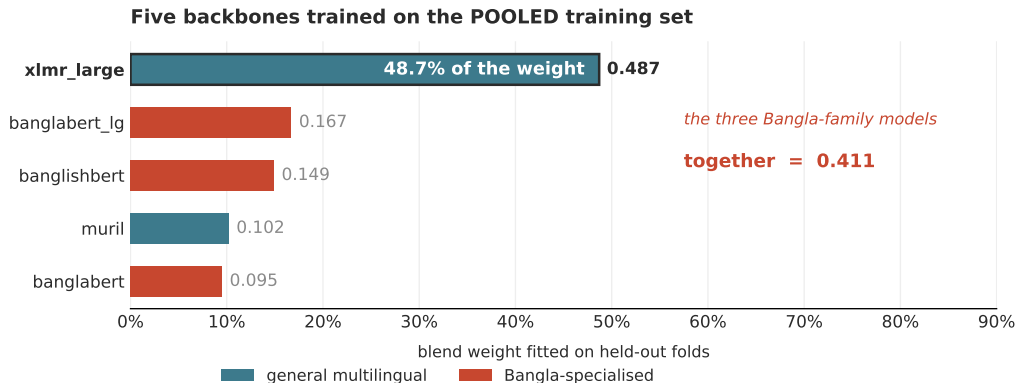


Every backbone trained on the **whole, unsplit** corpus, same folds, same head.

Finding 1 — one boundary is 71% of all errors

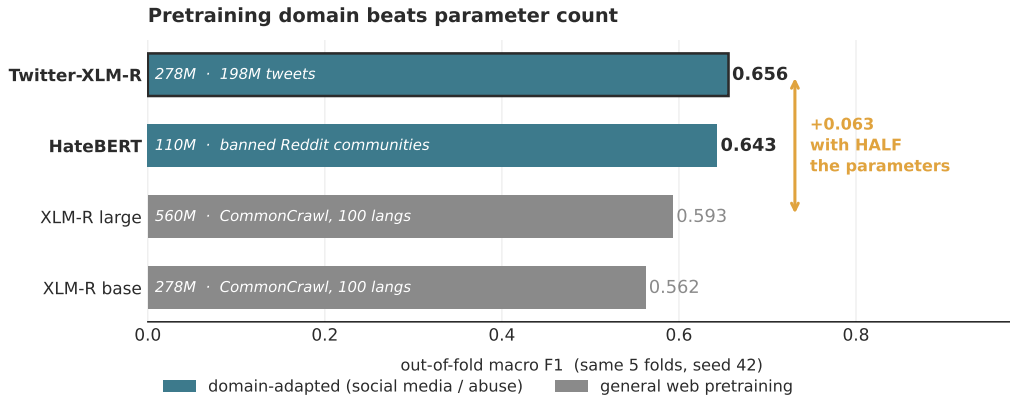


Finding 2 — one pooled model under-uses half the data



Five backbones, each fine-tuned on the **whole, unsplit** training set, then blended with weights fitted on held-out folds:

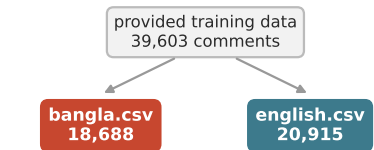
Finding 3 — domain of pretraining beats size of model



Same folds, same head, same pooled training set — only the pretrained backbone changes. A 278M model pretrained on tweets beats a 560M model pretrained on the open web.

Final system (1/2) — route by language, not by alphabet

Training data routed by **LANGUAGE**, not script



Bangla script + 2,080
romanised Banglish rows

BanglaBERT
BanglishBERT
BanglaBERT-large

English
(Latin script)

HateBERT
Twitter-XLM-R
XLM-R-large

Banglish follows the language, not the alphabet

...because their label priors are different

Bangla route
(18,688)

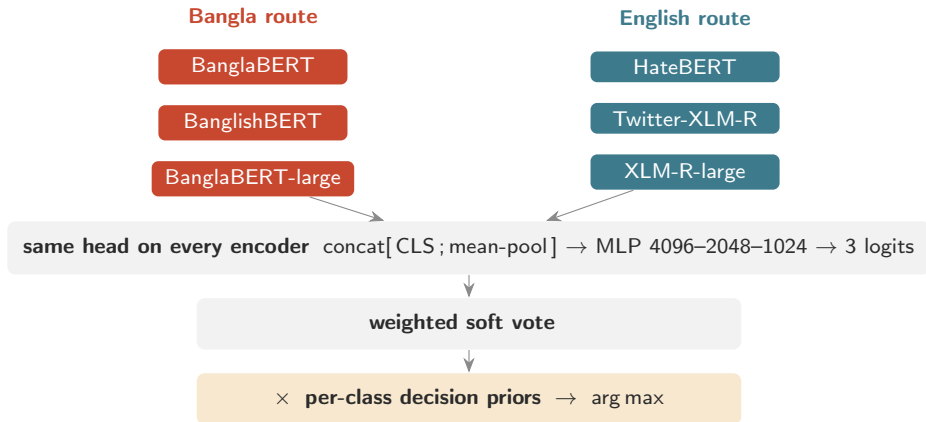


English route
(20,915)



Explicit (0) Subtle (1) Non-toxic (2)

Final system (2/2) — six encoders, one head, one calibrated decision



What we tried that did not work

Approach	Macro F1	Why we stopped
Qwen3-4B, few-shot prompting	0.502	Below every fine-tuned encoder
Qwen3-35B, few-shot prompting	0.550	8× the parameters, +0.048
Qwen3-4B, QLoRA fine-tune	—	Did not finish inside the competition window

Thank you